Name _____ Date _____

Activity Page 1.1 Use with Lesson 1.

Energy Scavenger Hunt

Change is all around you! **Energy** causes changes. Explore the space you are in. Look for things that change in some way. (Hint: Almost everything can!)

Find three examples of things that change. Complete the table to describe your choices.

Three samples have been done for you.

Something That Changes	How Does It Change?
a light bulb	It changes from not bright to bright.
an electric fan	It changes from not moving to moving.
the grass outside	It changes from short to tall.

Write two questions that you have about your examples.

1. _____

_____

2. _____

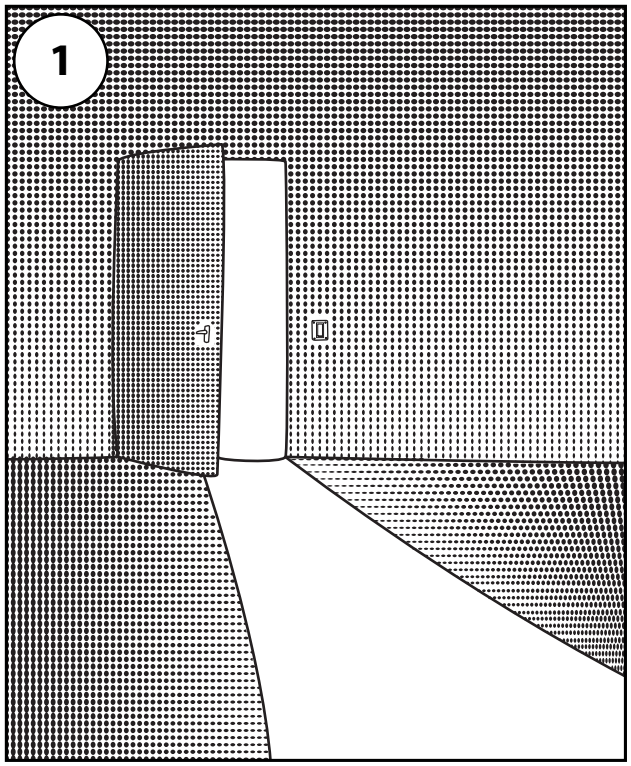
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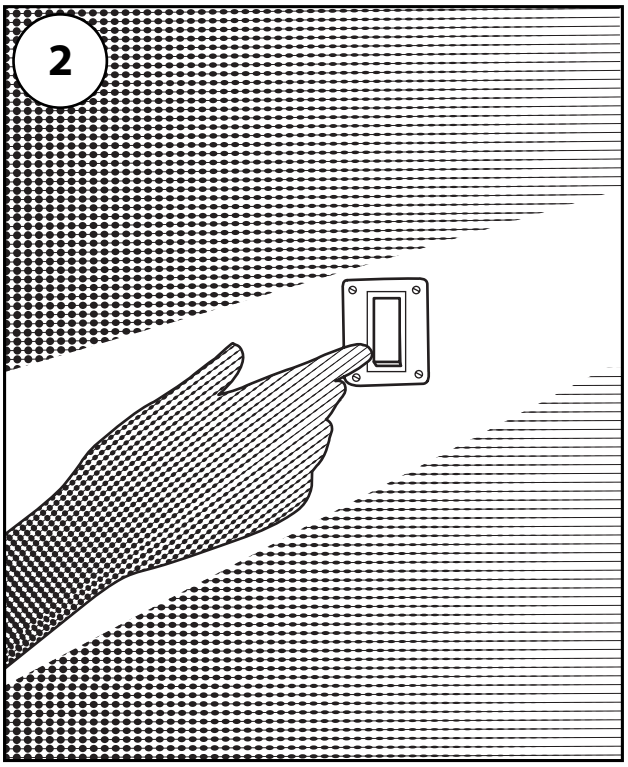
Name _____ Date _____

Activity Page 2.1 Use with Lesson 2.

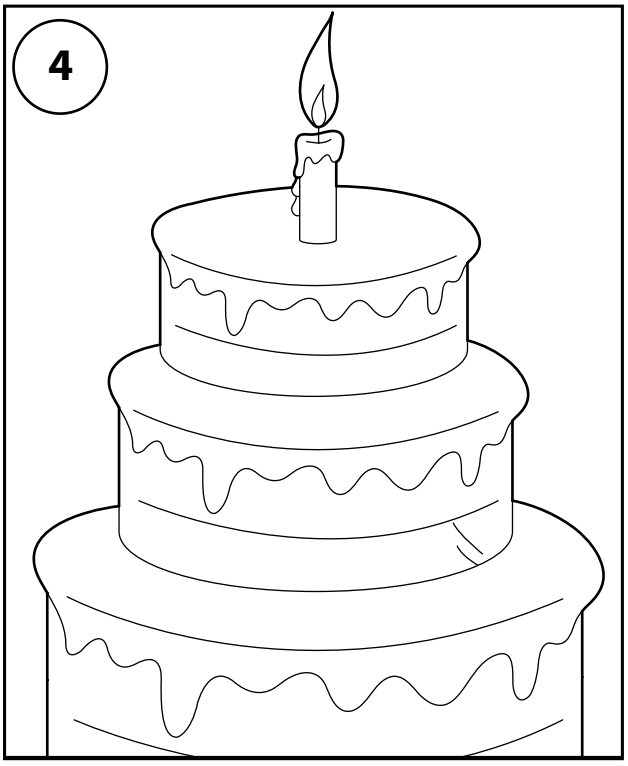
Surprise!

What changes occur in the party pictured here?









Name _____ Date _____

Activity Page 2.2 Use with Lesson 2.

Energy Causes Change

light	sound	electrical energy	motion energy	stored energy

<u>Choose</u> one form of energy from the box above. Circle your choice.

What change can this form of energy cause?

_____

_____

<u>Draw</u> a "before and after" cartoon to show a change, and indicate the form of energy that causes the change.

Before	After

<u>Explain</u> the cause and effect shown in your "before and after" cartoon.

What causes the change? What changes? (In other words, what is the cause, and what is the effect?)

_____

_____

_____

Name _____ Date _____

Activity Page 3.1 Use with Lesson 3.

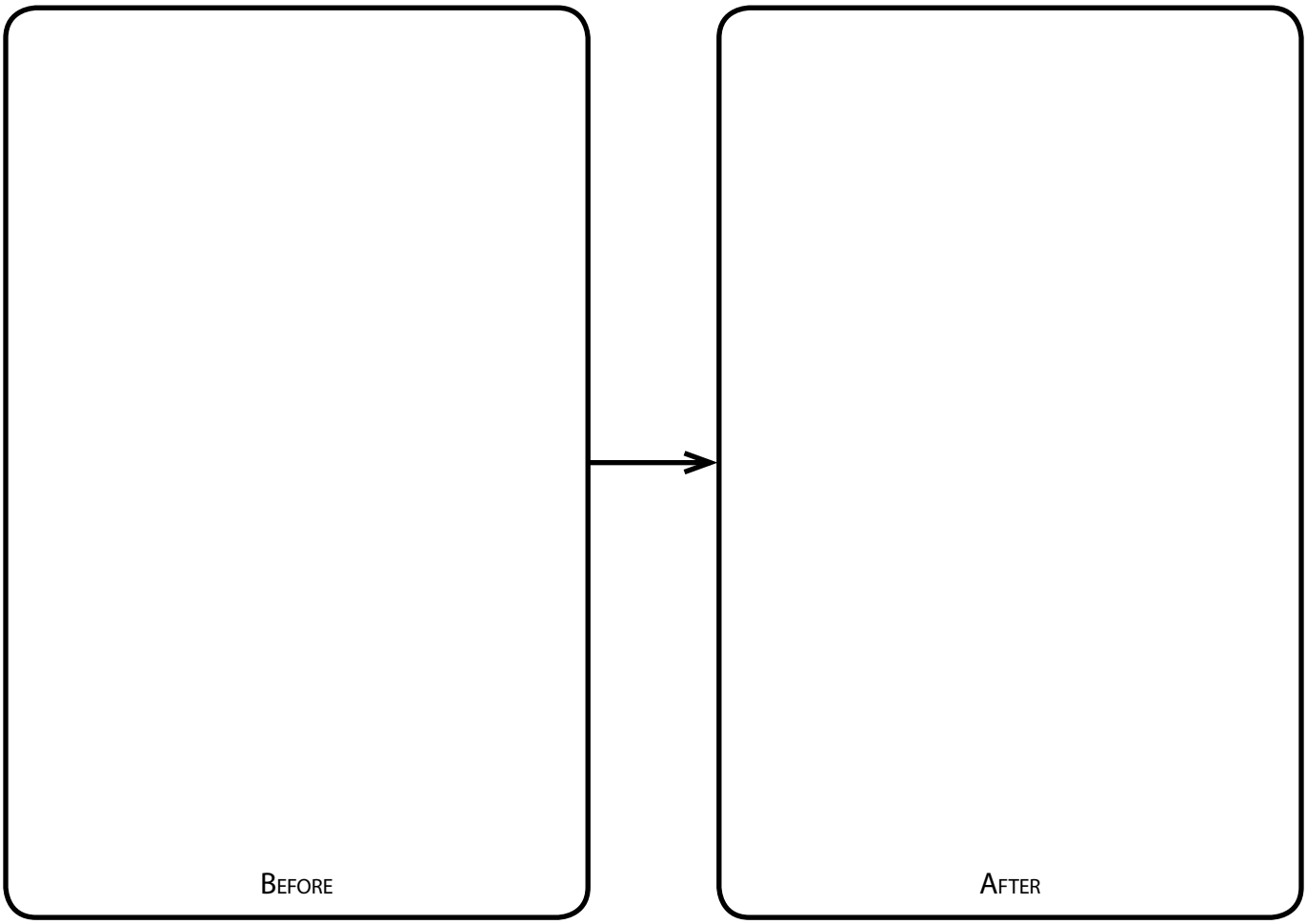
On the Move

Things are on the move everywhere that you look. Those moving objects can cause changes.

Identify an object that you know moves often.

_____

Draw before and after diagrams to show how the motion of this object can cause a change.



Write descriptions to explain your model.

_____ _____

_____ _____

_____ _____

Name _____ Date _____

Activity Page 4.1 Use with Lesson 4.

Ramp It Up (Day 1)

Different objects are moving at different **speeds** all around you! From the cars on the road to balls on the playground, all moving objects have speed and energy.

Is there a connection between speed and energy? You are about to find out!

STEP 1: Build a low ramp from cardboard propped at one end by a book. Place the cup a short distance from the bottom of the ramp (about the length of your foot).

STEP 2: Trial 1 Place your ball at the top of the ramp. Release the ball so it rolls down the ramp into the cup. Notice how far the ball causes the cup to move. Place a marker to show how far the cup moved.

STEP 3: Add books to make your ramp steeper so your ball will roll from a greater height. Place the cup in the same position in front of the ramp as you did before.

STEP 4: Trial 2 Place your ball at the top of the ramp. Release the ball so it rolls down the ramp into the cup. Notice how far the ball causes the cup to move.

STEP 5: Use your observations to complete the table.

	Trial 1	**Trial 2**
In which trial was the ramp height **higher**? In which trial was it **lower**?		
In which trial did the ball roll with **greater speed**? In which trial did it roll with **lesser speed**?		
In which trial did the ball move the cup **more**? In which trial did it move **less**?		
In which trial did the rolling ball have **more energy**? In which trial did it have **less energy**?		

STEP 6: Summarize your conclusion. Describe the relationship between energy and speed in your investigation. Use evidence to support your explanation.

Name _____ Date _____

Activity Page 4.2 Use with Lesson 4.

Drop It (Day 2)

Dropping an object from different heights produces different falling speeds for the object.

Does the relationship between speed and energy remain the same as what you observed when rolling a ball down a ramp?

STEP 1: Smooth the sand in the box so that there are no dents in the surface of the sand.

STEP 2: Trial 1 Hold the ball above the sand near one end of the box (not in the middle). Hold the ball at about knee-height.

STEP 3: Drop the ball into the sand. Gently pick the ball out of the sand without disrupting the sand. Notice the size of the crater that the ball's impact made in the sand.

STEP 4: Trial 2 Hold the ball above the sand near the opposite end of the box. Hold the ball at about waist-height.

STEP 5: Drop the ball into the sand again. Gently pick the ball out of the sand without disrupting the sand. Notice the size of the crater that the ball's impact made in the sand this time.

STEP 6: Use your observations to complete the table.

	Trial 1	**Trial 2**
In which trial was the ball dropped from **higher** position? In which trial was it dropped from a **lower** position?		
In which trial did the ball drop with **greater speed**? In which trial did it drop with **lesser speed**?		
In which trial did the ball have a **greater impact** on the sand? In which trial did it have **less** impact?		
In which trial did the falling ball have **more energy**? In which trial did it have **less energy**?		

STEP 7: Summarize your conclusion. Describe the relationship between energy and speed in your investigation. Use evidence to support your explanation.

Name _____ Date _____

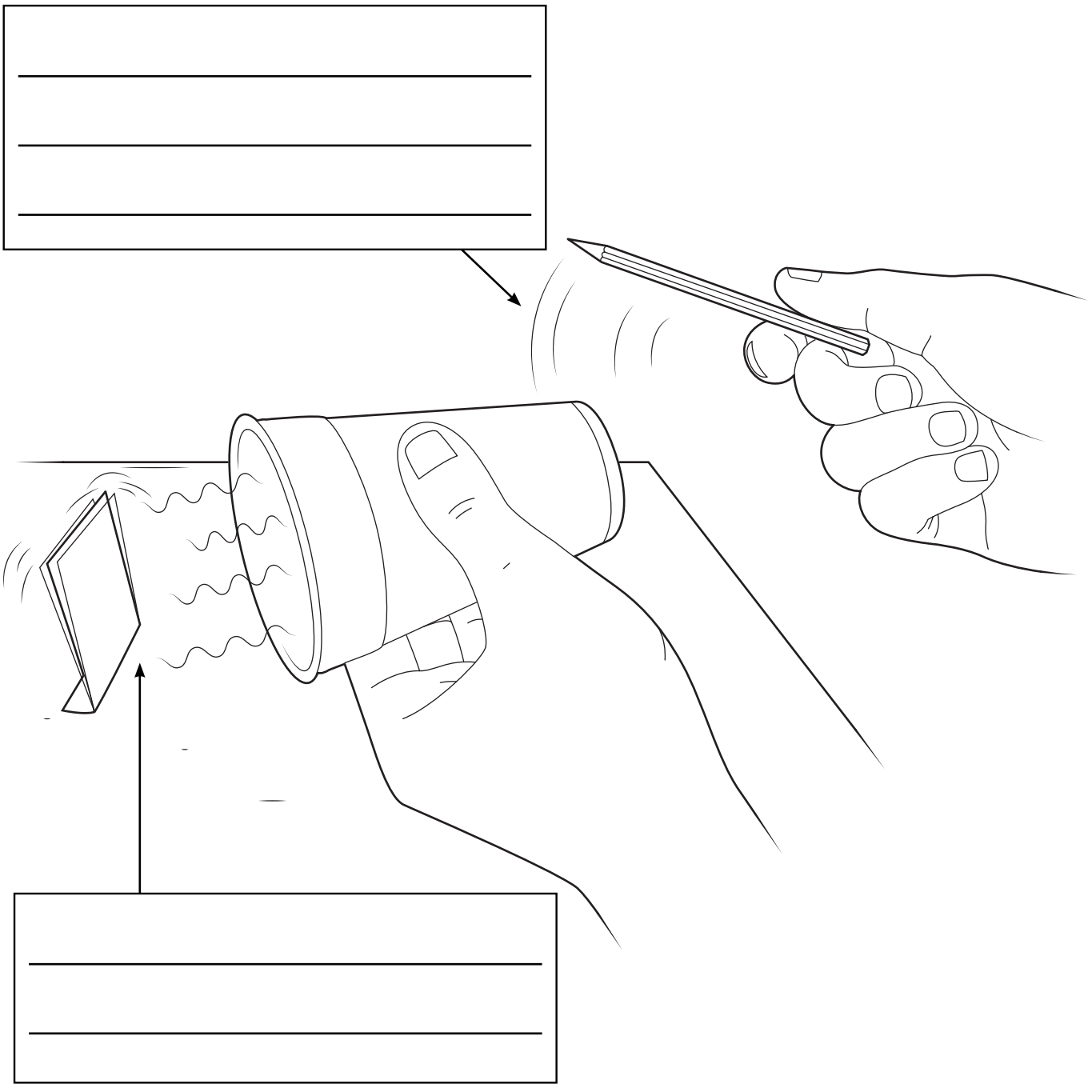
Activity Page 6.1 Use with Lesson 6.

Lesson 6 Check

<u>Write</u> captions on the picture to explain what is happening.

Use each of these words at least one time in your captions.

sound	energy	evidence	transfer



Name _____ Date _____

Activity Page 7.1 Use with Lesson 7.

Lesson 7 Check

Answer the items below to show what you have learned.

1. Which of the following is **evidence** of energy transfer? Circle all that apply.

a) a ball sitting at the top of a ramp

b) the sound from a vibrating tuning fork

c) a lighted wall across the room from a lamp

d) a crater produced by a meteorite impact

2. Draw a picture showing **energy transfer** from one object to another object by heat or electricity. Write a caption on the lines below to explain your drawing. Use the word ***evidence*** in your caption.

Name _____ Date _____

Investigating Energy Transfer

Station 1: Light Transfers Energy

STEP 1: Turn the flashlight on, and shine it at the wall so that its beam passes above the plate of chalk powder.

STEP 2: Gently dip the paintbrush bristles into the plate of chalk powder. Be careful to not make a mess!

STEP 3: Tap the brush gently with your free hand so the chalk powder falls back onto the plate. Watch as the chalk falls between the flashlight and the wall.

STEP 4: Describe what you see happen in terms of motion energy, chemical energy, electrical energy, and light energy.

STEP 5: Leave the station ready for the next group.

- Turn the flashlight off.
- Place the paintbrush back on the table.
- Be sure to let your teacher know if there is not enough chalk powder left on the plate. Other groups will need to repeat the same activity after you!

STEP 6: Answer the questions below about what you observed. Repeat the investigation if you need to.

1. Can you see a beam of light between the flashlight and the wall without the chalk dust?

2. How does the chalk dust change what you are able to observe?

3. What evidence did you observe that energy transferred from place to place?

Station 2: Heat Transfers Energy

SAFETY NOTE: Do not touch the hot plate or the beaker.

STEP 1: Use the tongs to place four ice cubes from the icebox/cooler into the beaker.

STEP 2: Watch what happens to the ice cubes.

STEP 3: Describe what you see happen in terms of electrical energy and heat energy.

STEP 4: When you are finished observing, let your teacher know to ready the station for the next group.

STEP 5: Answer the questions below about what you observed. Repeat the investigation if you need to.

1. What would happen if the temperature of the hot plate was decreased? How might this change the result?

2. Would the results change if you added more ice cubes? Why or why not?

3. What evidence did you observe that energy transferred from place to place at this station?

Station 3: Electricity Transfers Energy

SAFETY NOTE: Do not handle the surge protector power cord that is plugged into the wall electrical socket.

STEP 1: Observe the electronic device sitting on the table. Unplug the device from its charging cord. Notice what happens. Plug the charging cord back into the device. Notice what happens.

STEP 2: Trace the cord all the way to the surge protector. Switch the surge protector into the *off* position. Study the electronic device. Switch the surge protector back to the *on* position. Notice what changes.

STEP 3: Describe what you see happen in terms of electrical energy and stored energy.

STEP 4: Prepare the station for the next group.

- Plug the device back in (if it is unplugged).
- Turn the surge protector to the *on* position.
- Place the device on the table.

STEP 5: Answer the questions below, and repeat the investigation if you need to.

1. What causes a change in the device?

2. What happens when you turn the surge protector off? What happens when you turn it on?

3. What would happen if there were not a wire connecting the surge protector to the device?

4. What evidence did you observe that energy transferred from place to place at this station?

Name _____ Date _____

Activity Page 9.1 Use with Lesson 9.

Examples of Energy Transfer

You can observe examples of energy being transferred every day.

Identify and record some real-life examples in the table below.

Type of Energy	Description of Energy Transfer
Sound	
Light	
Heat	
Electricity	

Name _____ Date _____

Activity Page 9.2 (Page 1 of 2) Use with Lesson 9.

Claims, Evidence, and Reasoning About Energy Transfer

Part 1

Let's take a moment to review the important terms **claims**, **evidence**, and **reasoning**:

1. Claim:

What is a **claim**?

_____

_____

Why do people make **claims**?

_____

_____

2. Evidence:

What is **evidence**?

_____

_____

What are some types of **evidence**?

_____

_____

3. Reasoning:

Describe an example of scientific **reasoning**.

_____

_____

Why do you think people use **reasoning** in science?

_____

_____

Part 2

Now it's time to write a claim of your own.

1. Choose one example of energy transfer that you have learned about or investigated. Describe an example of how this form of energy transfers from one place to another.

2. What kind of evidence (data, observations, facts) was there that energy transfer took place?

3. Write a scientific claim about energy transfer. Follow this model: "I know _____ because_____ ."

Name _____ Date _____

Activity Page 10.1 Use with Lesson 10.

Lesson 10 Check





1. What two objects are about to collide in the picture above?

_____

_____

2. What form of energy can be observed before the collision happens?

_____

3. What forms of energy would you expect to observe as a result of this collision? Support your answer with evidence.

_____

_____

_____

_____

4. What two objects are about to collide in the picture above?

_____

_____

5. What form of energy can be observed before the collision happens?

_____

6. What forms of energy would you expect to observe as a result of this collision? Support your answer with evidence.

_____

_____

_____

_____

Name _____ Date _____

Activity Page 11.1 Use with Lesson 11.

Investigating Collisions (Day 1)

Predictions

Plan your investigation by completing the items below.

1. What do you want to find out? (Example: I want to find out what happens to energy when objects collide.)

2. What materials will you use for your investigation?

3. How will you make them collide?

4. Make a prediction. What do you think will happen when the objects collide?

5. What evidence might you observe that indicates energy transfers from one object to another?

Name _____ Date _____

Activity Page 11.2 (Page 1 of 2) Use with Lesson 11.

Investigating Collisions (Day 2)

Testing and Observation

Complete your investigation by completing the items below.

1. Use the materials you chose on Day 1, and build your ramp.

2. Conduct the investigation.

- Place the objects on the ramp.
- Roll the top objects down the ramp, and observe what happens.

3. Record your observations in the table below under Test 1.

Test	**Observations of Energy Transfer**
Test 1	
Test 2	

4. Change one variable (i.e., height or speed) in your design. Which variable will you change?

_____

5. Make a new prediction. What do you expect to see now that you changed this variable?

_____

_____

_____

6. Build your second design.

7. Perform the test again with the changed variable.

8. Record the new observations for Test 2 in the table.

9. Study the observations in the table. Answer the following questions:

- Was any energy transferred? Use evidence to support your answer.

- What are the differences between the two tests?

- Do you see any patterns? If so, what are they?

- What is the cause-and-effect relationship that you see? Write your answer as an *If, Then* statement.

  If _____ ,

  then _____

10. Evaluate the predictions you made before each test. Were your predictions accurate? Why or why not?

Name _____ Date _____

Activity Page 12.1 Use with Lesson 12.

Lesson 12 Check

Answer the questions to show what you know from this lesson.

1. What is the Big Question for this lesson?

_____

_____

2. Name a device that converts stored chemical energy of natural gas to heat energy to help people cook food.

_____

3. What energy transformation(s) takes place in a smoothie blender?

_____

_____

4. What problem does a wind-up kitchen timer solve? Use the word *solution,* as well as *want* or *need* (or both), in your answer.

_____

5. Describe two energy transformations that happen in a wind-up kitchen timer.

_____

_____

_____

6. How do people use a conversion from electrical energy to light energy in helpful ways in a kitchen or school classroom?

_____

_____

Name _____ Date _____

Activity Page 13.1 (Page 1 of 2) Use with Lesson 13.

Device Design Proposal (Day 1)

<u>Plan</u> a device to solve a problem using an energy transformation.

Problem to Be Solved: _____

_____

Device Name: _____

Energy Transformation:

Initial Form(s) of Energy	Final Form(s) of Energy
_____	_____
_____	_____

How will your team's device solve the problem by transforming (converting) energy?

_____

_____

_____

How will you know whether your device solves the problem or not?

_____

_____

Criteria: _____

Constraints (limitations): _____

Materials: Circle the materials your team will use. Write in any additional materials used on the lines below.

aluminum foil	cardboard tubes	cups	springs
ball	clear plastic wrap	glue	sticks
blocks	containers	ramps	string
boxes	craft paper	rubber bands	tape

_____

Name _____ Date _____

Activity Page 13.1 (Page 2 of 2) Use with Lesson 13.

Draw your device, and add labels to help show how your device will work.

Name _____ Date _____

Activity Page 13.2 Use with Lesson 13.

Device Test Results (Day 2)

Trial Test your device multiple times.	**Results** Briefly describe whether your device worked or not.	**Observations About Energy Transfer and Transformation** What evidence do you have that your device worked?	**Improvements or Fixes** How can you refine your device to work better in the future?
1.			
2.			
3.			

Make a claim about whether testing your device is important or not. Use your observations as evidence and reasoning to support your claim.

Name _____ Date _____

Activity Page 13.3 Use with Lesson 13.

Device Presentation Plan (Day 3)

Review your work on Activity Page 13.2 (Day 2). Summarize your work so far by answering these questions.

Problem to Be Solved: _____

_____

Device Name: _____

Energy Transformation:

Initial Form(s) of Energy	Final Form(s) of Energy
_____	_____
_____	_____

How well did your device solve the problem? _____

_____

_____

How did you build your device? _____

_____

_____

What were the results of your tests? _____

_____

_____

How did you refine your device? _____

_____

_____

Describe whether your revisions made your device work better or worse. Support your ideas by summarizing observations and evidence from each of your test trials.

_____

_____

Name _____ Date _____

Activity Page 13.4 Use with Lesson 13.

Device Presentation Scoring Guide (Day 4)

Use this guide as you listen to your classmates present their devices. For each line of the guide, give the team a score that reflects how well their presentation covered that section. For example, a score of 3 for "Defines the problem to be solved" would mean the presentation clearly told others of the problem to be solved.

Description	3	2	1
Defines the problem to be solved			
Meets the criteria defined for the device			
Meets the constraints or limitations defined for the device			
Identifies the initial and final forms of energy in the energy transformation			
Describes how the solution converts energy from one form to another to solve the problem			

I like your group's design because _____

_____

_____

What other questions would you ask about the device?

_____

_____

_____

Name _____ Date _____

Energy Vocabulary Crossword Puzzle

Use the words in the word bank to complete the crossword puzzle. Review the cards in your Core Vocabulary deck before you begin.

light	change	evidence	energy	motion	forms of energy*	position
sound	convert	heat	stored energy*	energy transfer*	electricity	
speed	energy of motion*	distance				

*Spaces that appear between words in terms above do not appear in the puzzle.

Across

2. when energy moves from one object to another or from one place to another

4. a type of energy that makes things hot

8. different types of energy, such as sound, heat, or light

11. the energy an object has while it is moving

12. to change

13. form of energy that powers lights in modern homes

14. how fast an object is moving

15. anything that helps prove or disprove an idea

16. to become different

Down

1. where something is in space

3. the process of an object changing position

5. a type of energy you can see with your eyes

6. the ability to cause change

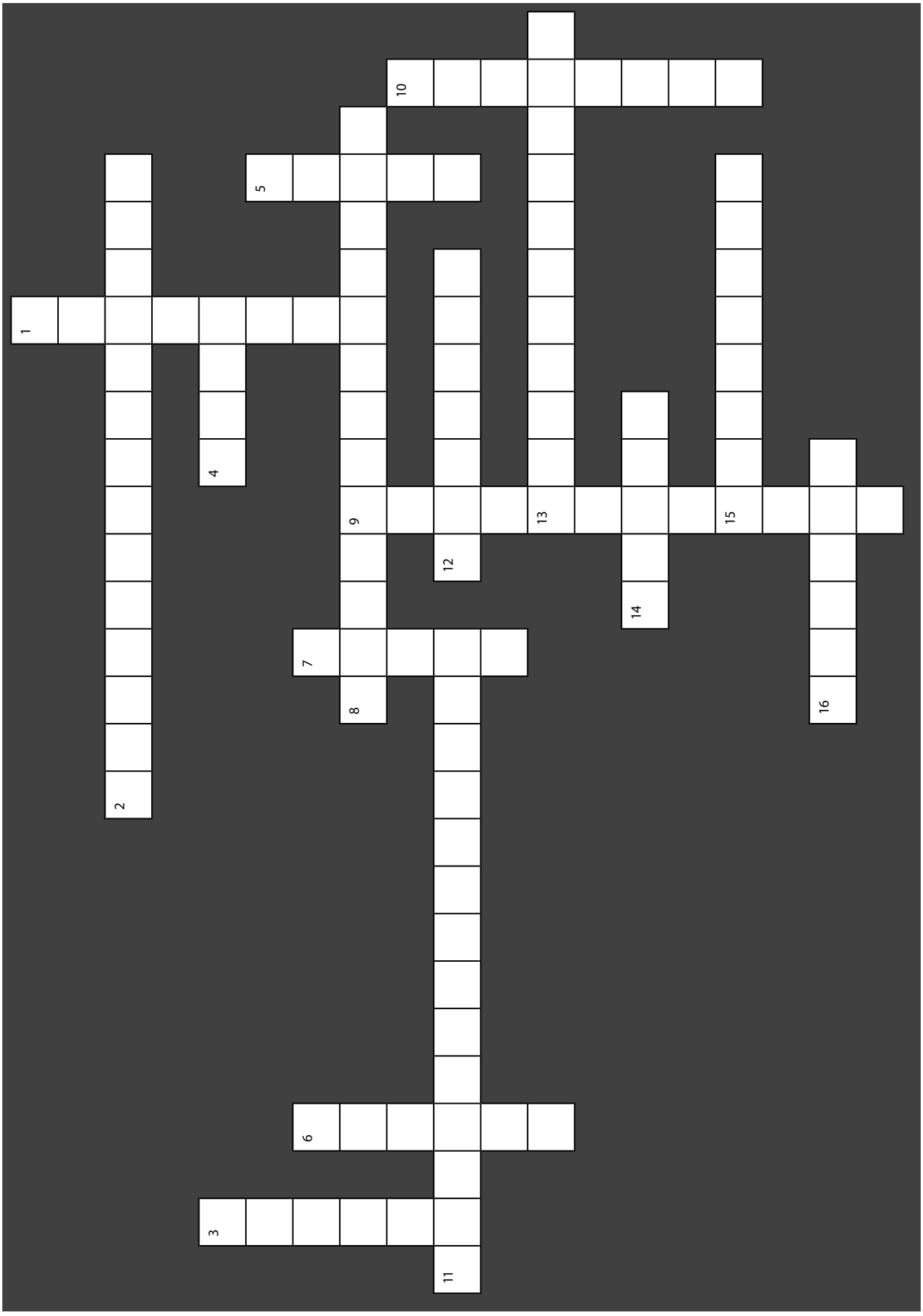
7. a type of energy you can hear with your ears

9. energy that has the potential to cause change at a later time

10. how far apart objects are

Name _____ Date _____

Energy Vocabulary Crossword Puzzle



Name _____ Date _____

Activity Page 13.6 Use with Lesson 13.

Energy Vocabulary Review

Complete each sentence with the correct Core Vocabulary term or phrase. Not every word in the word bank will be used, and some may be used more than once. Review the cards in your Core Vocabulary deck before you begin.

change	energy	forms of energy	stored energy	evidence	heat	light
motion	position	sound	transfer	collision	electricity	observation
model	speed	distance	energy of motion	convert	energy transformation	

1. When you add hot soup to a cold bowl, _____ energy transfers from the soup to the bowl.

2. If a book is falling off a shelf, the book has _____ because it is changing its _____.

3. Playing a drum produces _____, which is a form of energy that you can hear.

4. Energy can _____ from one object to another object.

5. When a bowling ball hits the bowling pins, a sound rings through the bowling alley. The change from motion energy to sound energy is an example of a(n) _____.

6. Energy is the ability to cause _____.

7. Heat, light, and sound are examples of _____.

8. A marble at the top of a ramp has _____ , because at any moment it can move down the ramp.

9. Computers are powered by _____.

10. _____ is a measure of how fast an object is moving, and _____ is a measure of how far apart objects are.

11. When light energy is converted to heat energy, this is known as _____.

Name _____ Date _____

Activity Page UR.1 Use with Unit Review.

Energy's Big Questions

The table below contains most of the Big Questions you have explored in the unit *Energy Transfer and Transformation*.

Circle the question that was assigned for you to answer.

Where can we observe evidence of energy causing change?	What evidence shows that energy is transferred from place to place?
What are some forms of energy?	How is energy involved in collisions?
How are energy, change, and movement of objects related?	What happens when objects collide?
How are energy and speed related?	How are collisions predictable?
What evidence shows that sound transfers energy from one place to another?	How do energy transformations help people?
What evidence shows that light, heat, and electricity transfer energy from place to place?	How did Thomas Edison use his knowledge of energy transfer and transformation to solve problems?

Write three or four sentences to answer the Big Question that was assigned to you and your partner.

Prepare a rough draft of your answer first on scrap paper, and then write your final draft below. Underline any terms in your answer which have a card in your Core Vocabulary deck.

Name _____ Date _____

Unit Assessment: What Have I Learned About Energy?

Answer the items below to show what you have learned.

1. Which sentences describe evidence of energy causing change? Circle the letter for each correct answer.

a) A basketball smashes into a backboard.

b) Electricity jumps from your fingertip to a doorknob, giving you a shock.

c) When the school bus starts up, you can hear the engine make a sound.

d) A stick of dynamite explodes.

2. Write the form or forms of energy described in each sentence below. Some sentences may describe more than one form of energy.

light	heat	motion	sound	electrical

a) The sun is shining on a nice summer day. _____

b) A firework explodes in the sky. _____

c) You kick a soccer ball at the goal. _____

d) You're happy and you know it, so you clap your hands. _____

e) You buckle your seat belt, and the school bus takes off to the school. _____

f) You turn on a battery-powered flashlight. _____

3. Draw a model of energy causing a change. Your drawing may show a before-and-after transformation, or you can use your imagination. Identify the type of energy causing the change. Write a sentence describing the change.

Type of energy: _____

Describe the change: _____

_____

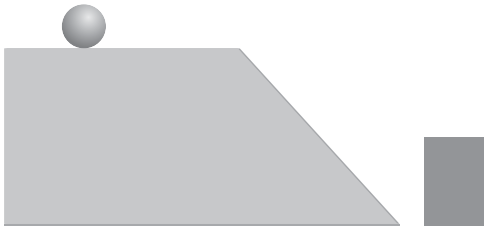
4. You are setting up an investigation to show the relationship between the speed of a moving object and its energy. You have a ball and a block. You also have three different types of ramps: a low-height ramp, a medium-height ramp, and a high-height ramp.



low



medium



high

Which ramp will result in the greatest amount of change? Explain your thinking using words from your Core Vocabulary deck.

5. Which ramp will result in the least amount of change? Explain why using words from your Core Vocabulary deck.

6. What is the relationship between the speed of an object and its energy of motion? Use evidence and examples you have learned during this unit to support your explanation.

7. Maria has been studying sound energy at school. She has written an explanation of sound using an example from her experience.

Maria says, "Sound energy only travels over long distances. When I am on the playground, I can hear the bell that rings from across the field that lets me know recess is over."

Do you agree with Maria? Use your own evidence to support your answer.

8. Which statements are evidence that energy has been transferred from object to object or place to place? Circle the letter for each correct answer.

a) Your friend kicks a goal while playing soccer.

b) Gravity pulls on an apple, but it doesn't fall out of a tree.

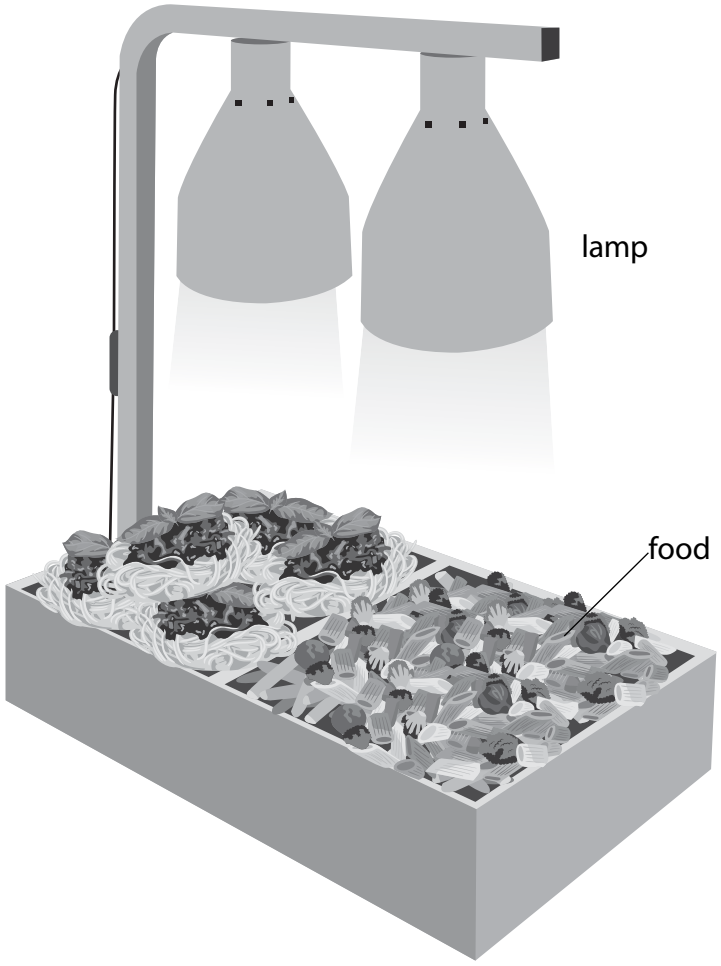
c) You push on the school door, but it stays closed.

d) You hear a bell ringing from far away.

e) You flip a light switch, and the room lights up.

f) A person pulls a wagon down a street.

g) The backpack you left in the car gets hot on a sunny day.



The school cafeteria keeps cooked food under lamps until students pick it up to eat. The top of each portion of food stays warmest.

9. What two kinds of energy are the lamps transferring to the food?

10. What evidence would support your claims about the types of energy used in the cafeteria example above? Use terms from your Core Vocabulary deck in your answer.

11. You roll a toy car down a ramp into a stack of four blocks. The top block falls off the stack.

You add some weight to the toy car and repeat the test, changing ONLY the weight of the car. This time, the top two blocks fall off the stack.

Based on this pattern, explain the result in terms of energy change.

What can you predict if even more weight is added to the toy car for the next test? Use terms from your Core Vocabulary deck in your answer.

12. Your teacher holds a ball up and drops it. She asks what happened to the energy of the ball. Here are the answers from four students.

Student A: The stored energy of the ball becomes motion energy.

Student B: The chemical energy of the ball becomes motion energy.

Student C: Heat transforms into motion energy.

Student D: Gravity is transferred into heat energy.

Which student is correct? Explain your reasoning using vocabulary, such as *position*, and examples that you have learned during this unit.

13. Your teacher is leaving over the weekend, but the house plants in his classroom can't be left without water or light. Describe a device that could solve this problem. The solution would need to automatically provide water **and** light to a plant when people are away.

14. If you were to build your device for your teacher, how would you test it to see if it works, and how might you improve your design based on your test results?

15. A student named Maria made a sketch of her solution to the same problem. Maria's solution includes plugging a lamp into a timer that works in a wall socket. At the same times each day, the timer switches the lamp on and off. Her solution also includes a container of water that allows water to drip slowly through a hose down into the planter. The container of water sits on a stack of books so it is positioned higher than the planter.

Complete the labels on Maria's diagram to identify the types of energy that are converted in her design.

Maria's drip hose solution converts

_____

to _____ .

Maria's lamp design solution converts

_____

to _____ .

